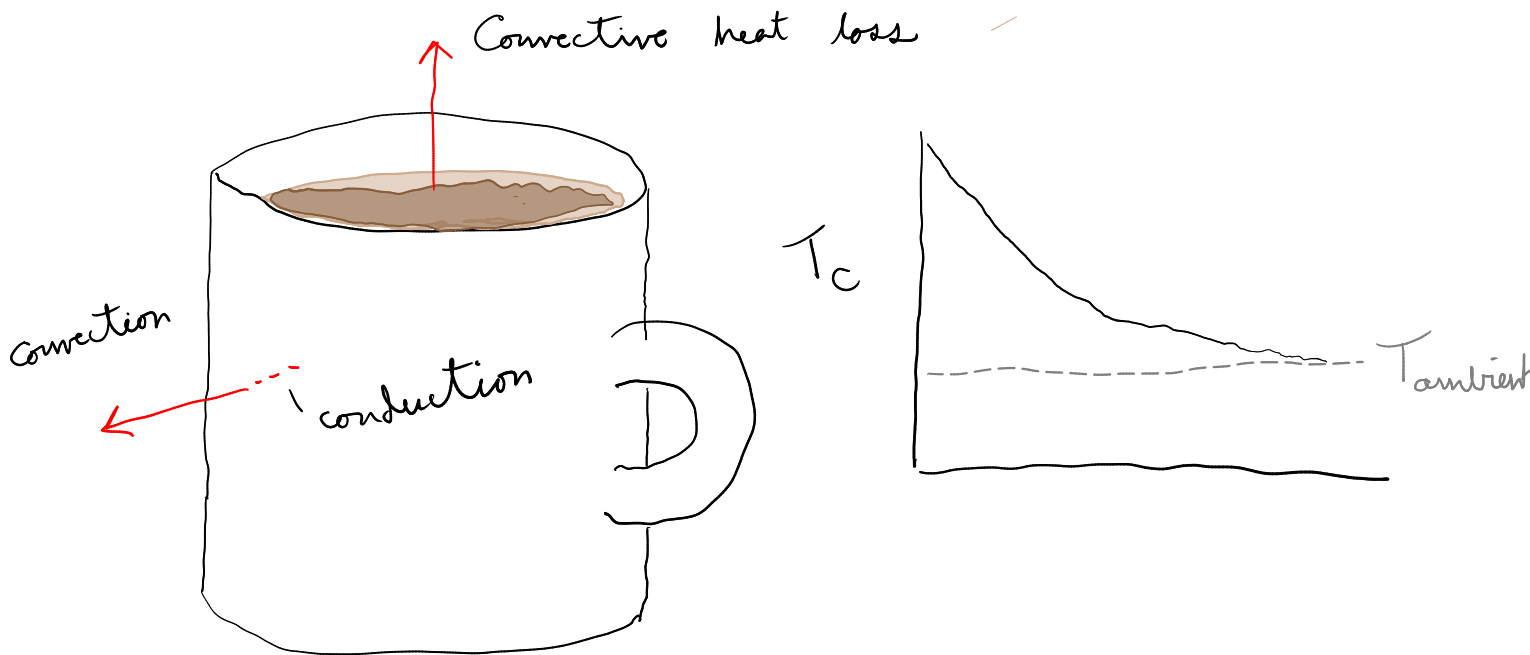


Cooling cup of coffee

The problem: Let's try to develop a model that can predict the coffee temperature change over time.

Sketches



State variable: Thermal energy in coffee

Q

Boundary: Outer surface of coffee or cup

Heat balance

$$\text{accumulation} = \cancel{\text{heat flow}_{\text{in}}} - \cancel{\text{heat flow}_{\text{out}}} + \cancel{\text{generation}}$$

↪

$$\text{accumulation} = - \text{heat flow out}$$

$$\frac{dQ}{dt} = - \dot{Q} \quad (W)$$

Newton's law of cooling

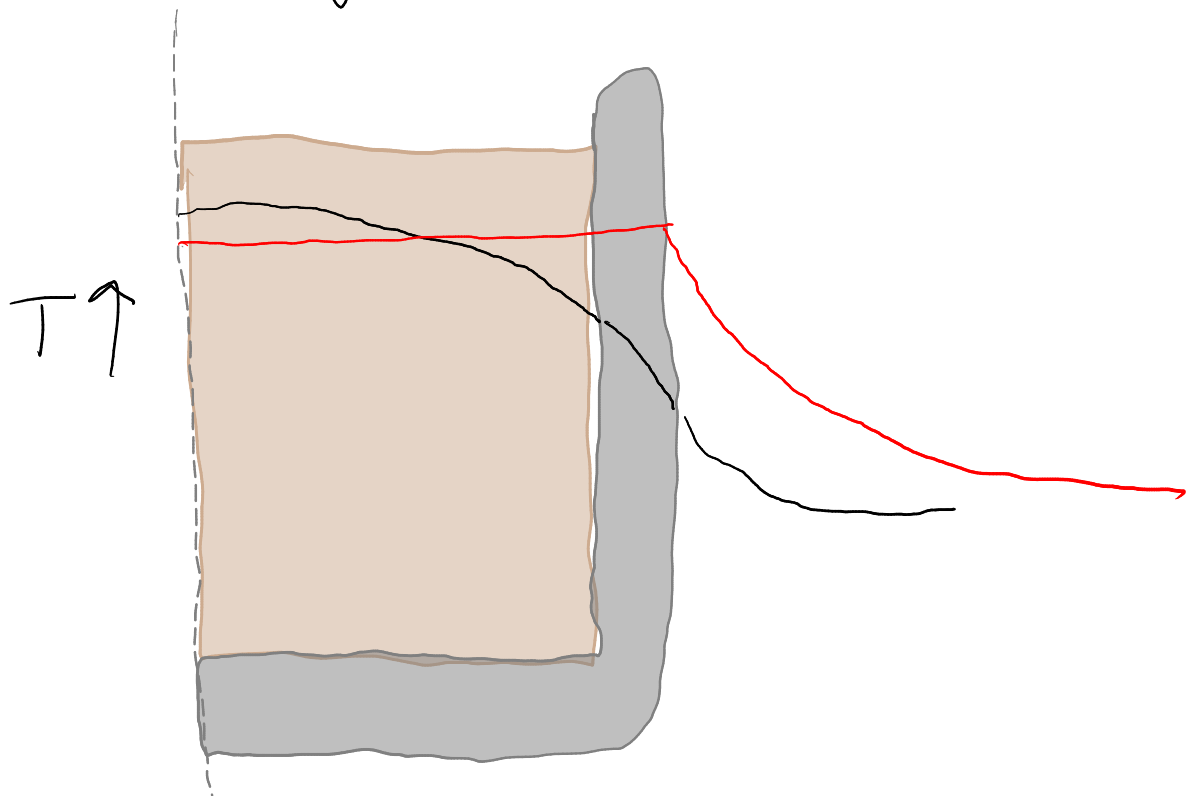
$$q = h (T_s - T_\infty) \quad (\text{W/m}^2)$$

or

$$\dot{Q} = h (T_s - T_\infty) A \quad (\text{W})$$

Let's go for a lumped capacitance approach (lumped parameter).

We will ignore internal resistance, and will lump any resistance from the mug into h .



Linking equations

Energy $\leftrightarrow$ T $\leftrightarrow$ heat loss

$$\frac{dQ}{dt} = c_p m \frac{dT}{dt}$$

$$m = \rho V$$

$$\dot{Q} = A q$$

surface area

heat flux

heat
flow rate

$$\frac{dQ}{dt} = -A \cdot h \cdot (T_c - T_\infty)$$

$$\frac{dT_c}{dt} = \frac{dQ}{dt} \cdot \frac{1}{c_p m}$$

$$\frac{dT_c}{dt} = - \frac{A \cdot h}{c_p \cdot m} \cdot (T_c - T_\infty) \quad \textcircled{1}$$

$$\frac{d\Theta}{dt} = -c \quad \textcircled{1}$$

$$\Theta(t) = \Theta_0 e^{-ct}$$

or

$$T_c(t) = (T_{c,0} - T_\infty) e^{-ct} + T_\infty$$

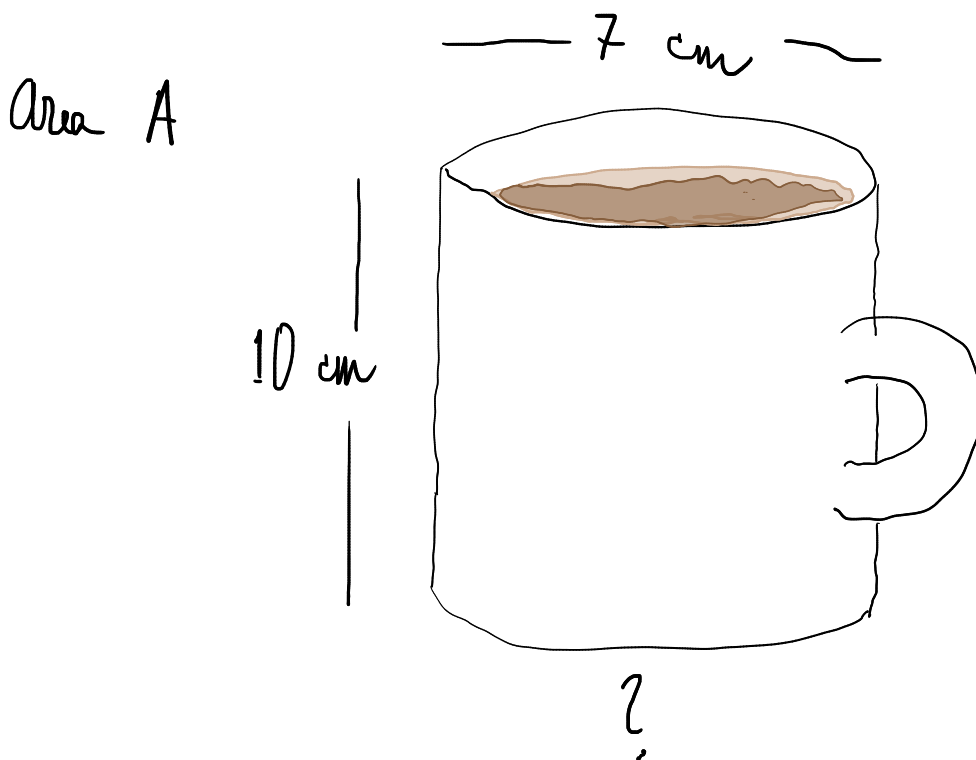
Parameter values and inputs

TABLE 1.1 Typical values of the convection heat transfer coefficient

Process	h (W/m ² · K)
Free convection	
Gases	2–25
Liquids	50–1000
Forced convection	
Gases	25–250
Liquids	100–20,000
Convection with phase change	
Boiling or condensation	2500–100,000

h from incropera

$$c_p = 4.2 \text{ J/g} \cdot \text{K}$$



$$\begin{aligned} A &= 7 \text{ cm} \pi \cdot 10 \text{ cm} + \pi (3.5 \text{ cm})^2 \\ &= 220 \text{ cm}^2 + 38 \text{ cm}^2 \\ &= 258 \text{ cm}^2 = 0.0258 \text{ m}^2 \end{aligned}$$

mass $m \approx 300 \text{ g}$